Food remedy Co. API

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Explainability and Transparency in Personalised Food Recommendation Systems

Introduction

Digital health and nutrition apps are becoming an important part of how people make food choices today. Many of these apps use personalised recommendation systems to suggest meals or products based on a user's preferences, goals, and past behaviour. While this makes things more convenient and tailored, it also creates a challenge. Most users do not fully understand how these recommendations are generated. In many cases, the system works like a "black box," where users can see the outcome but not the reasoning behind it.

This research explores how personalised food recommendation systems explain their suggestions to users. It focuses on the role of transparency and explainability in shaping user trust, confidence, and decision-making. In particular, it looks at how different explanation methods, such as labels, scoring systems, and descriptive insights, help users understand and interact with these recommendations.

Background and Literature Review

Recommender systems work by analysing user data, including preferences, habits, and past interactions, to generate personalised suggestions. In food and nutrition contexts, this helps users deal with complex information and make better dietary decisions. However, as these systems become more advanced, it becomes harder to understand how they actually make decisions. This lack of clarity raises concerns about transparency and accountability.

Explainable AI, often referred to as XAI, aims to make system decisions easier for users to understand. In recommendation systems, this means clearly showing why a particular food item or meal is suggested. Explanations are not only useful for improving usability, but they also play an important role in building trust. When users understand the reasoning behind a recommendation, they are more likely to accept it and feel confident in their choices.

Transparency is closely linked to trust. When systems provide clear and simple explanations, users feel more in control and are more willing to rely on them. On the other hand, when there is little or no explanation, users may feel unsure or even sceptical. Research suggests that transparency helps users gradually build confidence in how a system works over time.

Food recommendation systems use several methods to explain their suggestions. Some rely on nutritional labels that show calories, ingredients, or macronutrients. Others use scoring systems, such as colour-coded indicators or health ratings, to simplify complex information. Some applications also provide descriptive insights in plain language, explaining why something is recommended. Visual elements like charts or graphs are sometimes included as well. Each of these approaches offers different levels of clarity and usefulness, which influences how users interpret recommendations.

Research Methodology

This study uses a qualitative and comparative approach to examine how explainability is applied in personalised food recommendation systems. Data was collected by analysing nutrition-related applications and reviewing academic literature on explainable recommender systems. Visual examples of explanation features, including labels, scoring systems, and descriptive insights, were also examined to understand how systems communicate their reasoning.

The analysis was guided by key criteria such as clarity, transparency, usability, and trustworthiness. Clarity refers to how easily users can understand the explanations provided. Transparency measures how much information the system reveals about how decisions are made. Usability focuses on how simple it is for users to interact with explanation features. Trustworthiness reflects how confident users feel about the recommendations.

A comparative approach was used to identify common explanation techniques across different platforms, along with their strengths and limitations. This helped highlight patterns in how explainability is implemented and how it influences user behaviour and decision-making.

Analysis and Findings

The analysis shows that food recommendation systems mainly rely on three types of explanations. These include label-based explanations, scoring systems, and descriptive insights. Label-based explanations provide detailed nutritional information such as calorie counts and ingredient lists. These offer a high level of transparency, but they often require some prior knowledge to understand properly.

Scoring systems simplify complex nutritional information into formats such as letter grades, numerical ratings, or colour-coded indicators. These are very effective for quick decision-making, especially for users who may not have much time or expertise. However, this simplicity can reduce transparency because some details are left out.

Descriptive insights provide explanations in natural language and directly connect recommendations to user goals or needs. For example, a system might recommend a product because it is high in protein or low in sugar. These explanations make the system feel more engaging and easier to understand, but they can sometimes lack depth or miss important details.

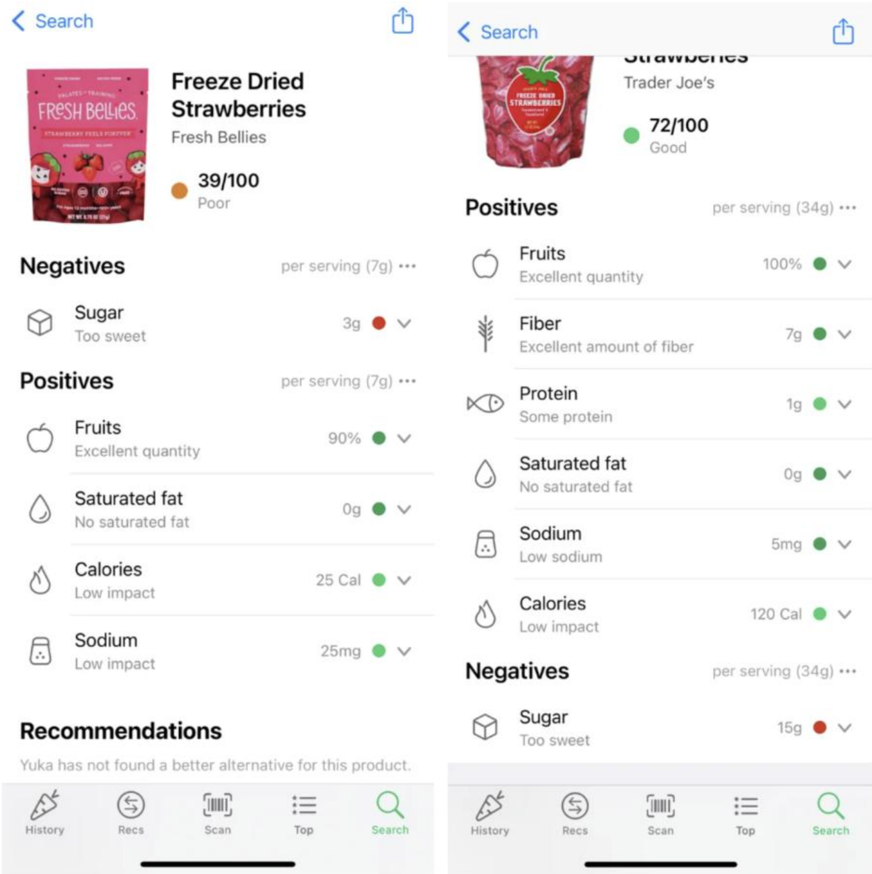
One of the key findings of this research is the trade-off between simplicity and transparency. Systems that provide detailed explanations are more transparent but can feel overwhelming. Simpler systems are easier to use but may hide important information. The most effective systems combine different explanation methods to balance these two aspects.

The study also found that explainability has a strong impact on user trust and decision-making. When users understand why a recommendation is made, they are more likely to accept it. Transparent systems help users form a clearer understanding of how recommendations work, which increases confidence and satisfaction. On the other hand, when explanations are missing, users may hesitate or ignore the recommendations.

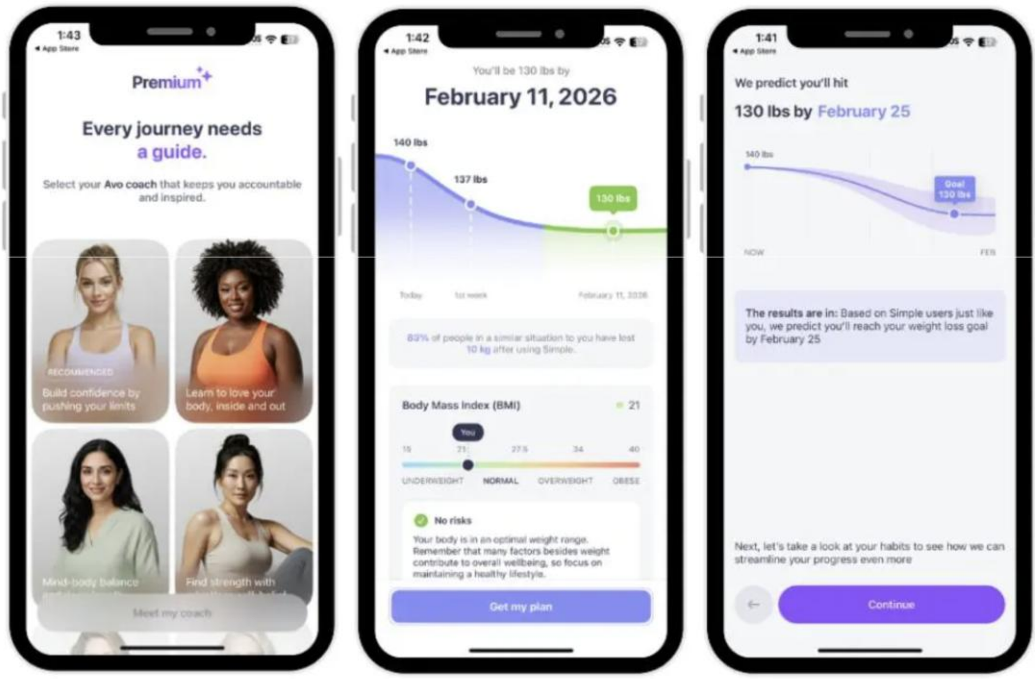
Discussion

The findings show that explainability is an important part of user experience design in food recommendation systems. Explanations need to be clear, relevant, and easy to understand. Using simple language, layered information, and visual elements can improve both usability and engagement.

At the same time, there are challenges in implementing explainability. These include balancing simplicity with detail, avoiding misleading explanations, and ensuring consistency across recommendations. Future improvements may include interactive features that allow users to explore why something was recommended, as well as personalised explanation styles based on individual preferences. Advances in explainable AI will be important in addressing these challenges and improving transparency.



Yuka gives recommendations on better food alternatives. Image source: https://mamaknowsnutrition.com/yuka-app-reviews/



Simple App has an AI assistant that can answer questions and give suggestions for physical workouts and recipes. Image source: https://fortune.com/article/best-nutrition-apps/

Image source: https://adamosoft.com/blog/healthcare-software-development/personalized-nutrition-app/

Conclusion

This research shows that explainability and transparency are essential for the success of personalised food recommendation systems. When users understand the reasoning behind recommendations, they are more likely to trust the system and make confident decisions. While each explanation method has its strengths, the most effective systems combine multiple approaches to balance usability and transparency. As AI continues to develop in the health and nutrition space, focusing on explainability will be key to creating systems that are both reliable and user-friendly.

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